Response to Reviewers

ARES: Autonomous Real-Time Evaluation System

Submission ID: eda35bea-a77d-4e7d-ba30-edb38c688296

Dear Editor and Reviewers,
We sincerely thank the Editor and all three reviewers for their detailed and constructive feedback. The comments were extremely valuable in identifying issues related to formatting, reproducibility, empirical rigor, and presentation clarity.
We have carefully revised the manuscript (now titled **ARES_LP.pdf**) and implemented all suggested corrections. In addition to fixing the formatting and rendering issues, we substantially improved the manuscript's scientific rigor, added formal mathematical foundations, performed statistical testing and ablation studies, and provided a public proof-of-concept repository for reproducibility.

Below we provide a detailed, point-by-point response to each reviewer comment.

Reviewer 1

Comment 1: Unreadable mathematical formulas and formatting issues

"Unreadable mathematical formulas where text overflows; PDF/DOC formatting problems."

Response: Completely fixed.

All mathematical expressions were re-typeset into properly rendered, numbered equations. The manuscript now contains a dedicated **Decision-Theoretic Formulation** section with formal definitions, equations, and a proposition. Previously overflowing equations such as loss functions, throughput definitions, and verification operators are now clearly formatted and readable.

Comment 2: Tables containing raw LaTeX code

"Tables contain raw LaTeX code instead of rendered output."

Response: Fixed.

All tables were rebuilt and re-rendered. Benchmarking tables, capability comparison tables, and dataset tables are now fully readable with descriptive captions.

Comment 3: Confusing/overlaid plots

"Plots with confusing and overlaid visual components."

Response: Fixed and simplified.

All figures were redrawn. Overlays were removed, axes were clarified, and captions were expanded to describe experimental conditions.

Comment 4: Reproducibility and access to benchmarks/models

"Provide access to benchmarks and open-source the code."

Response: Implemented.

We created and linked a public proof-of-concept repository:

`https://github.com/gurarpitzz/ARES_POC`

The manuscript now includes step-by-step reproduction commands, file-to-equation mapping, and instructions for running verification and evaluation scripts.

Reviewer 2

Comment 1: Unresolved citation markers "[?]"

Response: All placeholders removed. References were completed and verified. In-text citations now correctly map to the bibliography.

Comment 2: Author name formatting issues

Response: Corrected. The author block, affiliation, and metadata now conform to journal standards.

Comment 3: Garbled code in figures and tables

Response: All affected figures and tables were replaced with correctly rendered versions. Captions now clearly describe the data and axes.

Comment 4: Title too general

Response: Clarified.

The title now includes the subtitle:

An AI-Driven Multi-Modal Truth Infrastructure for Misinformation Defense

The abstract and introduction were rewritten to emphasize technical contributions.

Reviewer 3

Comment 1: Limited empirical support and no statistical testing

Response: Substantially expanded.

We added:

- $K = 5$ independent runs (mean $\pm$ std)
- Paired t-test
- Wilcoxon signed-rank test
- Cohen's d effect size
- Reported $p < 0.01$ for benchmark comparisons

Comment 2: Superficial training details

Response: Expanded.

We now include full hyperparameters, optimizer configuration (AdamW), learning rate schedule, adversarial loss, regularization terms, batch size, and epoch details.

Comment 3: Internal metrics without external validation

Response: Addressed.

A new section on **External Validation and Distribution Shift** formalizes internal vs external datasets and includes stress-test methodology. The GitHub PoC includes sample FEVER/LIAR evaluation scripts.

Comment 4: Need ablation studies

Response: Implemented.

We added a **Module-wise Ablation and Counterfactual Analysis** section evaluating ARES$-X$ for each module. Results are presented with mean $\pm$ std across runs.

Comment 5: Promotional narrative tone

Response: Adjusted.

Narrative sections were relocated to a clearly labeled background subsection. The conclusion and abstract were rewritten to maintain a scientific tone.

Additional Major Changes Made

Beyond reviewer comments, we added:

- Formal decision-theoretic verification framework with proposition and proof sketch
- Algorithms and pseudocode mapped to code modules
- Computational complexity analysis
- Dataset table with sources and volumes
- Ethical governance and misuse prevention section
- Limitations and future work section
- Reworked all figures and tables
- UI-to-mathematical trace mapping

Reproducibility Guide (included in manuscript)

```
git clone https://github.com/gurarpitzz/ARES_POC.git
cd ARES_POC
pip install -r requirements.txt
```

```
python main.py "The Earth orbits the Sun" --k 10 --m 5
python evaluate.py --dataset liar --samples 50

uvicorn app:app --reload
```

Summary

In summary, the revised manuscript corrects all formatting issues, improves scientific rigor, adds statistical validation, includes ablation studies, formalizes the theoretical foundation, and provides a public reproducible implementation. We believe the manuscript is now significantly strengthened and addresses all reviewer concerns.

We sincerely thank the reviewers for their insightful feedback and hope the revised version meets the standards of the journal.

Sincerely,
Gurarpit Singh